A calibrated battery of cheap weak-class tests on the NIST prime curves' b

2026-10-22

Status

One native run; no ECDLP cost measured. Protocol: PROTOCOL.md, frozen in commit `b06037d99` before the real curves were scored. Run: run/, produced by `target/release/b_weak_class_battery --real` at that commit (clean tree, rustc 1.97.0, 4 threads, 933 s wall; `run/run.json`). The pre-flight estimate in the protocol was 35 minutes; the run took 15.6.

This is not an index-calculus measurement. It has no rho ratio, no phase is priced and no scoreboard row is owed (`AGENTS.md` §7a). It tests hypothesis 1 of research/p224_hypothesis_ranking_20261008/ (a weak class recognisable from `(p, a, b)`, found by grinding seeds) with a wider family of statistics than the single KS test of `b mod q`.

Weak-class battery: real NIST curves against 8000 seed-generated nulls per curve

Left: empirical p-value of the real curve (darker = smaller). Right: fraction of 100 planted instances detected at the corrected threshold 5.88e-4.
Grey cell: positive control failed, so the real-curve result is inconclusive. Flag rule: p at most 5.88e-4 (Bonferroni, 85 tests).

	P-192	P-224	P-256	P-384	P-521	P-192	P-224	P-256	P-384	P-521
jac_b (A)	0.323	0.420	0.248	0.654	0.082	1.00	1.00	1.00	1.00	1.00
jac_c (A)	0.784	0.532	1.000	0.647	0.422	1.00	1.00	1.00	1.00	1.00
jac_j (A)	0.530	0.786	0.788	0.326	0.536	1.00	1.00	1.00	1.00	1.00
jac_disc (A)	0.321	0.418	0.927	0.323	0.412	1.00	1.00	1.00	1.00	1.00
depth_b (B)	0.497	1.000	0.198	0.496	0.770	0.00	1.00	0.00	0.00	1.00
depth_disc (B)	1.000	0.414	0.444	1.000	0.645	0.00	1.00	0.00	0.00	1.00
hw_b (C)	0.722	0.546	0.488	0.578	1.000	1.00	1.00	1.00	1.00	1.00
run_b (C)	0.523	0.193	0.986	0.955	0.875	1.00	1.00	1.00	1.00	1.00
naf_b (C)	0.520	0.715	0.433	0.643	0.752	1.00	1.00	1.00	1.00	1.00
hw_c (C)	0.044	1.000	0.796	0.297	0.348	0.99	1.00	1.00	1.00	1.00
run_c (C)	0.516	0.835	0.212	0.086	0.063	1.00	1.00	1.00	1.00	1.00
smooth_b (D)	0.118	0.802	0.777	0.813	0.117	1.00	1.00	1.00	1.00	1.00
smooth_c (D)	0.073	0.307	0.034	0.448	0.927	1.00	1.00	1.00	1.00	1.00
smooth_j (D)	1.000	1.000	1.000	0.305	0.747	1.00	1.00	1.00	1.00	1.00
hw_j (E)	0.709	0.643	0.742	0.168	0.931	1.00	1.00	1.00	1.00	1.00
run_j (E)	0.790	0.100	0.629	0.787	0.986	1.00	1.00	1.00	1.00	1.00
d3 (F)	1.000	0.063	0.598	1.000	1.000	0.00	0.00	0.00	0.00	0.00

Real curves scored. Smallest p the panel can give: 1.25e-4. Source: summary.json and plants_*.csv in this directory.
Statistics are oriented so large is suspicious; p = (1 + #(null at least as large)) / (N + 1). Family letters: A Jacobi, B power-residue depth, C bits, D smoothness, E j bits, F 3-isogeny.

Left: the real curves' empirical p-values. Right: planted-instance detection. 8000 nulls per curve, 100 plants per cell, commit b06037d99, run/summary.json.

Claim classes (`controlled-experiment` §1): every sentence about a curve below is **comparative/statistical**. The unit-tested algebra (division polynomial, Jacobi, NAF identity) is **exact computation**. The label **DERIVED** marks one consistency argument; **OPEN** marks what is not established.

Result

Observation (MEASURED, `run/summary.json`).

- **No curve × statistic is flagged.** The pre-declared rule is Bonferroni over $m = 5 \times 17 = 85$ tests, $\alpha = 0.05$, flag iff $p \le 5.88 \times 10^{-4}$ (at most 3 of 8000 nulls at or above the curve). The smallest real-curve p is 0.034 (P-256, `smooth_c`), 57 times the threshold.
- **Positive controls passed in 74 of 85 cells.** In every passing cell at least 99 of 100 planted instances were detected at the corrected threshold. **11 cells are inconclusive** (grey in the figure): `d3` on all five curves, and `depth_b` and `depth_disc` on P-192, P-256 and P-384. Their real-curve values are reported but are not evidence either way.
- **Calibration gate passed.** In all 85 cells, 0 to 62 of 1000 held-out null draws score $p \le 0.05$ (gate: at most 70; counts below 30 are ties on coarse discrete statistics and are conservative, not a failure). The mean number of seeds tried per accepted null draw is 9.6 to 12.1.
- **Two of the 74 conclusive cells have $p \le 0.05$** (P-192 `hw_c`, 0.044; P-256 `smooth_c`, 0.034). Under a continuous null about 3.7 would be expected; the discrete statistics make the count conservative.

Real-curve empirical p-values

`p = (1 + #{null ≥ x}) / 8001`, upper tail. † marks a cell whose positive control failed (inconclusive). All 85 rows with values and null intervals are in the appendix.

statistic	P-192	P-224	P-256	P-384	P-521
jac_b	0.323	0.42	0.248	0.654	0.082
jac_c	0.784	0.532	1	0.647	0.422
jac_j	0.53	0.786	0.788	0.326	0.536
jac_disc	0.321	0.418	0.927	0.323	0.412
depth_b	0.497 †	1	0.198 †	0.496 †	0.77
depth_disc	1 †	0.414	0.444 †	1 †	0.645
hw_b	0.722	0.546	0.488	0.578	1
run_b	0.523	0.193	0.986	0.955	0.875
naf_b	0.52	0.715	0.433	0.643	0.752
hw_c	0.044	1	0.796	0.297	0.348
run_c	0.516	0.835	0.212	0.086	0.063
smooth_b	0.118	0.802	0.777	0.813	0.117
smooth_c	0.073	0.307	0.034	0.448	0.927
smooth_j	1	1	1	0.305	0.747
hw_j	0.709	0.643	0.742	0.168	0.931
run_j	0.79	0.1	0.629	0.787	0.986
d3	1 †	0.063 †	0.598 †	1 †	1 †

Why the 11 cells are inconclusive

These are limits of the detector, not of the curves.

- **`d3`** (rational 3-isogeny kernels): its support under the conditioning is `{0, 1, 2, 4}`, and the largest value has about 6% null mass, so no value can reach the corrected threshold (the best attainable `p` on any curve is 0.055 to 0.077 in the dry run, 0.063 for P-224's value here). On P-192 and P-384, `p ≡ 2 mod 3` and `d3` is 0 for every null draw. Predicted in the protocol (§4).
- **`depth_b`, `depth_disc`**: the power-residue depth is capped by the small-prime part `G` of `p − 1`. P-192 and P-384 have `v₂(p − 1) = 1` and no other small factor, so the support is `{0, 1}`; `depth_disc` cannot be planted there (`p ≡ 2 mod 3`). **P-256 was not predicted in the protocol:** `v₂(p − 1) = 1` there as well, so `G = 2550 = 2·3·5²·17` (`log2 G = 11.32`) and the planted maximum (11.32) is reached by 5 of 8000 nulls (`p = 6/8001`), just above the threshold. Disclosed here; the rule (inconclusive) was applied as written.

Observations outside the pre-registered rule

Exploratory, not part of the Bonferroni family, not findings.

1. **`smooth_j` is exactly 0 on P-192, P-224 and P-256.** The integer `j` has no prime factor below `2^16` on three of the five curves (P-384: 19.5 bits; P-521: 6.3 bits). The pre-registered direction for smoothness is the upper tail, so these cells are `p = 1`. The null frequency of `smooth_j = 0` is 0.046 to 0.056 per curve (`run/nulls_*.csv.gz`), so three or more of five has probability about `10 × 0.05³ ≈ 1.2 × 10^-3` by hand arithmetic. It was noticed after the run, among 17 statistics and both tails, so it carries a look-elsewhere factor of at least 3 (three smoothness cells) and in fact more. No mechanism is known, and it is not confirmed. A confirmation needs a new frozen protocol and an independent set of curves generated by the same procedure (for example the other X9.62 and SEC 2 seeded prime curves).
2. **`d3 = 4` on P-224** (all four 3-division x-coordinates rational, none with a rational point). **DERIVED:** with no rational 3-torsion point, this says Frobenius acts as the scalar −1 on `E[3]`, i.e. `3` divides the conductor of `Z[π]`. The earlier measured P-224 conductor `f = 3` (`research/p224_hypothesis_ranking_20261008/run/cm_discriminant.json`) predicts exactly this, and P-256 (conductor 1) gives `d3 = 1`. So `d3 = 4` is a re-expression of a known fact, not new structure.

3. **Torsion carries no information on these curves.** The curves have prime order, so there is no rational 2-torsion or 3-torsion point and no rational 2-isogeny. The unit test confirms this for all five curves (the conditioning accepts them). A battery that reports “number of rational 2-/3-torsion points” would report 0 for every real curve and every conditioned null.

Interpretation (labelled, not a finding)

Within these seventeen statistics, on these five curves, nothing is more extreme than the corrected threshold where the detector demonstrably works. That narrows hypothesis 1 only to classes that stay inside the null on all 74 conclusive cells. Hypothesis 1’s grind scenario was already closed beyond about 30 bits of selection by the seed provenance work (`research/nist_seed_provenance_20261016/`); this battery adds no weight to that and takes none away.

Method in brief

`PROTOCOL.md` is the specification. In short:

- **Null (negative control).** 8000 draws per curve: random 160-bit seed, `c = x962_c_from_seed`, `b` a square root of `−27/c` with a seed-derived sign, conditioned on odd order and no rational 3-torsion (cheap necessary conditions of the prime order every NIST curve has). The same code scores the null, the held-out draws, the plants and the real curves.
- **Statistics.** `jac_*` (Jacobi pattern of `b`, `c`, `j`, $4a^3 + 27b^2$ over 32 odd primes), `depth_*` (joint power-residue depth), `hw_*`, `run_*`, `naf_b` (bit structure and signed-digit weight), `smooth_*` (smoothness below `2^16`), `d3` (3-isogeny pattern from the division polynomial). All are computed from `(p, a, b)` and `c` without counting points.
- **Positive controls.** 100 constructed instances per cell (Chinese remaindering, products of small primes, `r^g`, bit patterns), through the same conditioning and statistic code.
- **Multiple testing.** Bonferroni over 85 tests, declared in the protocol.

Power limits

A null here does not show that no weak class exists.

- The instrument sees a class only if its members move one of the 17 statistics beyond the top 4 of 8000 null draws. The planted classes show what that takes: a quadratic-residue pattern on all 32 primes, a 40-bit run, a 24-digit NAF, a fully 2^16-smooth integer, a depth far above the null maximum. A subtler bias, for example “slightly more residues than expected on the first 12 primes”, is not claimed to be visible. No power curve in selection bits was measured.
- A class defined by the group order, the trace, the CM field, an isogeny degree of 5 or more, or any other function of `b` outside these families is invisible. The 3-isogeny statistic has no power at the corrected level.
- The depth statistics have no power on P-192, P-256 and P-384, whose `p − 1` has no useful small-prime part.
- The null is conditioned only on odd order and no rational 3-torsion, not on prime order itself, and uses a seed-derived sign that the real curves may not follow. A statistic correlated with 5-torsion or higher would be mis-calibrated, and none of the 17 is.
- The statistics within a curve are dependent. Bonferroni is conservative.
- Five curves, one real instance each: there is no replication on a second set of real curves.

Checklist (`controlled-experiment` §6)

- [x] every published sentence classified (§1 above)
- [x] negative control: seed-generated null, conditioned, 8000 per curve, plus 1000 held-out draws
- [x] positive control on the same path for every cell; 11 cells failed it and are reported inconclusive
- [x] one instrument and identical budgets across arms
- [x] power check `2/(N+1) < α/m`: $2.5 \times 10^{-4} < 5.9 \times 10^{-4}$; $N + 1 = 8001 \geq 6800$
- [x] pre-flight runtime estimate (35 min) against actual (15.6 min)
- [x] `PROTOCOL.md` frozen (commit `b06037d99`) before the real run; dry run on scratch output with real scoring off
- [x] raw rows committed (`run/*.csv.gz`)
- [x] decision rule and correction applied as written
- [x] verdict wording fixed; scope sentence present

- [ ] superseded claims: none superseded. The predecessor README's "1c: statistical probes on `b` are the right kind of test: partly" stays as written; this study widens the probes and does not change that status

Reproduce

```
cargo test --release --lib cryptanalysis::b_weak_class_battery
cargo build --release --bin b_weak_class_battery
target/release/b_weak_class_battery --real \
  --out research/b_weak_class_battery_20261022/run
```

The run is deterministic (seed labels are fixed); `run.json` holds the wall time and differs between runs. The bin writes plain CSV; the committed files are those CSVs compressed with `gzip -9` (their hashes below include the gzip header time). SHA-256 of the committed outputs:

```
bcfc0b1eac6845e78b553d6ad51b8060262ceafef4427e0234508f63358fe7c9 summary.json
b651bb9e9973b6dbe7ef807ca8f20c1a476e81872b568126d1f06dafa7500feb battery.svg
3f594b7feff1d872c45a985c49b5ab726e15143f2cff80b25b1b6df002e96a0e holdout_P-192.csv.gz
1d6e84826739566c0091ec3836eaa9b7ea14afb6f22b902176722cfd720c81bb holdout_P-224.csv.gz
e953e92eb2ebba98ccf617d33b0306f0d7945528fbae240868a9eca95f9bbeee holdout_P-256.csv.gz
2674f82f1cd3211a01d2c81e7c3f517acd543eca1ce1663ba180de9ac2713b68 holdout_P-384.csv.gz
1ef7862994c469a628e397be1689247abef96919b29f7997f6b8b12e0be7689e holdout_P-521.csv.gz
18fa62613ecd204b11b997fe09668724e730efa48942db7bb3aef35b6d0bbc7d nulls_P-192.csv.gz
8572ec3d1f6d894842e7af37ae438b4d7493a9336ab3404b007693cceff40415 nulls_P-224.csv.gz
3343e88666628ed11757d049e741f21d4cedb9348378a5d3fcae9050ac032825 nulls_P-256.csv.gz
2566784afb3d19cc42d3be26f26a4fb76748c295bc33ccbb9a5e4723ca41f3c7 nulls_P-384.csv.gz
8aa1bd6d43b02ed8b5f36ef6376be2e94ada117d5669611b88cb1c1a27789058 nulls_P-521.csv.gz
b9e4fc67bd62473f7f6ea296384367bc60d28d6c59c783ace2efd0135b41cde3 plants_P-192.csv.gz
20f1100d9733e2e6e5bc6d51aec8819a768e8fbba7aa97dcb4cf899b89f6ffab plants_P-224.csv.gz
75dfd4eff3cc2b9d47840656b6efd98f4478a12686276a9948bf833db85a62d9 plants_P-256.csv.gz
6b4d49a9f424ae39db688b886c6345ca240faa727555c66278cb07ac5e5735ca plants_P-384.csv.gz
571e779dba55c12f42eb9aa241341445710fedea2412a23ab519029f6ede83ea plants_P-521.csv.gz
```

Appendix: all 85 cells

Columns: curve, statistic, real value, null median [2.5%, 97.5%], null maximum, empirical `p`, planted instances detected at the corrected threshold, held-out null draws with `p ≤ 0.05` (of 1000), verdict (fixed wording; the nothing-flagged verdict applies to cells whose positive control passed).

curve	statistic	value	null median [2.5%, 97.5%]	null max	p	plants detected	held-out p ≤ 0.05	verdict
P-192	jac_b	1.06066	0.707107 [0.0, 2.12132]	3.712311	0.3231	100/100	46	within the null distribution
P-192	jac_c	0.353553	0.707107 [0.0, 2.12132]	3.889087	0.7843	100/100	42	within the null distribution
P-192	jac_j	0.707107	0.707107 [0.0, 2.12132]	4.065864	0.5302	100/100	28	within the null distribution
P-192	jac_disc	1.06066	0.707107 [0.0, 2.12132]	3.535534	0.321	100/100	46	within the null distribution
P-192	depth_b	1.0	0.0 [0.0, 1.0]	1.0	0.4971	0/100	0	positive control failed: inconclusive
P-192	depth_disc	0.0	0.0 [0.0, 0.0]	0.0	1	0/100	0	positive control failed: inconclusive
P-192	hw_b	0.433013	0.721688 [0.0, 2.165064]	3.752777	0.7219	100/100	56	within the null distribution

curve	statistic	value	null median [2.5%, 97.5%]	null max	p	plants detected	held-out p ≤ 0.05	verdict
P-192	run_b	8.0	8.0 [5.0, 12.0]	19.0	0.5228	100/100	42	within the null distribution
P-192	naf_b	0.795495	0.795495 [0.0, 2.12132]	4.507806	0.5197	100/100	59	within the null distribution
P-192	hw_c	2.098366	0.651217 [0.072357, 2.243081]	3.97966	0.0444	99/100	35	within the null distribution
P-192	run_c	8.0	8.0 [5.0, 12.0]	23.0	0.5163	100/100	42	within the null distribution
P-192	smooth_b	29.594285	13.40833 [0.0, 41.806084]	76.215891	0.1182	100/100	54	within the null distribution
P-192	smooth_c	33.463855	13.266054 [0.0, 42.297326]	95.467412	0.0734	100/100	53	within the null distribution
P-192	smooth_j	0.0	13.612062 [0.0, 41.474014]	79.979169	1	100/100	54	within the null distribution
P-192	hw_j	0.433013	0.721688 [0.0, 2.165064]	4.041452	0.7089	100/100	47	within the null distribution
P-192	run_j	7.0	8.0 [5.0, 12.0]	21.0	0.7903	100/100	39	within the null distribution
P-192	d3	0.0	0.0 [0.0, 0.0]	0.0	1	0/100	0	positive control failed: inconclusive
P-224	jac_b	0.883883	0.707107 [0.0, 2.12132]	4.065864	0.4203	100/100	25	within the null distribution
P-224	jac_c	0.707107	0.707107 [0.0, 2.12132]	4.242641	0.5321	100/100	48	within the null distribution
P-224	jac_j	0.353553	0.707107 [0.0, 2.298097]	3.889087	0.7862	100/100	34	within the null distribution
P-224	jac_disc	0.883883	0.707107 [0.0, 2.12132]	3.889087	0.4181	100/100	27	within the null distribution
P-224	depth_b	0.0	2.0 [0.0, 7.321928]	14.994353	1	100/100	47	within the null distribution
P-224	depth_disc	3.321928	2.584963 [1.0, 8.087463]	13.906891	0.4139	100/100	46	within the null distribution
P-224	hw_b	0.668153	0.668153 [0.0, 2.271721]	3.741657	0.5458	100/100	42	within the null distribution
P-224	run_b	10.0	8.0 [5.0, 13.0]	20.0	0.1931	100/100	30	within the null distribution
P-224	naf_b	0.409159	0.654654 [0.081832, 2.291288]	4.091585	0.7147	100/100	34	within the null distribution
P-224	hw_c	0.066965	0.602685 [0.066965, 2.209843]	3.950932	1	100/100	39	within the null distribution

curve	statistic	value	null median [2.5%, 97.5%]	null max	p	plants detected	held-out p ≤ 0.05	verdict
P-224	run_c	7.0	8.0 [6.0, 13.0]	21.0	0.8351	100/100	27	within the null distribution
P-224	smooth_b	4.807355	13.476113 [0.0, 42.019111]	78.058859	0.8015	100/100	49	within the null distribution
P-224	smooth_c	19.591166	13.56403 [0.0, 41.97487]	71.958417	0.3066	100/100	50	within the null distribution
P-224	smooth_j	0.0	13.331337 [0.0, 40.871857]	85.274789	1	100/100	55	within the null distribution
P-224	hw_j	0.534522	0.668153 [0.0, 2.13809]	4.008919	0.6432	100/100	47	within the null distribution
P-224	run_j	11.0	8.0 [6.0, 12.0]	21.0	0.0996	100/100	50	within the null distribution
P-224	d3	4.0	1.0 [0.0, 4.0]	4.0	0.0627	0/100	0	positive control failed: inconclusive
P-256	jac_b	1.237437	0.707107 [0.0, 2.12132]	3.712311	0.248	100/100	43	within the null distribution
P-256	jac_c	0.0	0.707107 [0.0, 2.12132]	3.358757	1	100/100	40	within the null distribution
P-256	jac_j	0.353553	0.707107 [0.0, 2.12132]	3.712311	0.7877	100/100	28	within the null distribution
P-256	jac_disc	0.176777	0.707107 [0.0, 2.12132]	3.712311	0.9274	100/100	37	within the null distribution
P-256	depth_b	3.321928	1.0 [0.0, 6.228819]	11.316282	0.1976	0/100	42	positive control failed: inconclusive
P-256	depth_disc	2.584963	1.0 [1.0, 6.672425]	11.316282	0.4439	0/100	39	positive control failed: inconclusive
P-256	hw_b	0.75	0.625 [0.0, 2.25]	3.875	0.4883	100/100	51	within the null distribution
P-256	run_b	6.0	8.0 [6.0, 13.0]	22.0	0.9855	100/100	32	within the null distribution
P-256	naf_b	0.842012	0.612372 [0.076547, 2.21985]	4.056967	0.4333	100/100	58	within the null distribution
P-256	hw_c	0.313112	0.688847 [0.062622, 2.31703]	3.569478	0.7957	100/100	37	within the null distribution
P-256	run_c	10.0	8.0 [6.0, 13.0]	22.0	0.2125	100/100	32	within the null distribution
P-256	smooth_b	5.61471	13.420881 [0.0, 42.027406]	82.108625	0.7774	100/100	38	within the null distribution

curve	statistic	value	null median [2.5%, 97.5%]	null max	p	plants detected	held-out p ≤ 0.05	verdict
P-256	smooth_c	39.498831	13.38869 [0.0, 41.720796]	79.962971	0.0336	100/100	44	within the null distribution
P-256	smooth_j	0.0	13.273359 [0.0, 42.123963]	73.9768	1	100/100	57	within the null distribution
P-256	hw_j	0.375	0.625 [0.0, 2.25]	4.0	0.7419	100/100	50	within the null distribution
P-256	run_j	8.0	8.0 [6.0, 13.0]	21.0	0.6292	100/100	33	within the null distribution
P-256	d3	1.0	1.0 [0.0, 4.0]	4.0	0.5981	0/100	0	positive control failed: inconclusive
P-384	jac_b	0.53033	0.707107 [0.0, 2.12132]	3.535534	0.654	100/100	38	within the null distribution
P-384	jac_c	0.53033	0.707107 [0.0, 2.12132]	4.065864	0.6468	100/100	37	within the null distribution
P-384	jac_j	1.06066	0.707107 [0.0, 2.12132]	3.889087	0.3261	100/100	36	within the null distribution
P-384	jac_disc	1.06066	0.707107 [0.0, 2.12132]	3.535534	0.3235	100/100	46	within the null distribution
P-384	depth_b	1.0	0.0 [0.0, 1.0]	1.0	0.4956	0/100	0	positive control failed: inconclusive
P-384	depth_disc	0.0	0.0 [0.0, 0.0]	0.0	1	0/100	0	positive control failed: inconclusive
P-384	hw_b	0.612372	0.714435 [0.0, 2.245366]	4.184545	0.5782	100/100	45	within the null distribution
P-384	run_b	7.0	9.0 [6.0, 13.0]	20.0	0.9548	100/100	47	within the null distribution
P-384	naf_b	0.5625	0.75 [0.0, 2.25]	4.6875	0.6433	100/100	48	within the null distribution
P-384	hw_c	1.07305	0.664269 [0.051098, 2.299393]	3.627931	0.2968	100/100	36	within the null distribution
P-384	run_c	12.0	9.0 [6.0, 13.0]	20.0	0.0865	100/100	53	within the null distribution
P-384	smooth_b	4.643856	13.419829 [0.0, 42.630203]	75.861938	0.8128	100/100	51	within the null distribution
P-384	smooth_c	14.42213	12.953105 [0.0, 41.357992]	79.080088	0.4484	100/100	62	within the null distribution
P-384	smooth_j	19.544491	13.497353 [0.0, 42.143493]	80.758487	0.3052	100/100	52	within the null distribution

curve	statistic	value	null median [2.5%, 97.5%]	null max	p	plants detected	held-out p ≤ 0.05	verdict
P-384	hw_j	1.428869	0.714435 [0.0, 2.245366]	3.878359	0.168	100/100	36	within the null distribution
P-384	run_j	8.0	9.0 [6.0, 13.0]	22.0	0.7874	100/100	45	within the null distribution
P-384	d3	0.0	0.0 [0.0, 0.0]	0.0	1	0/100	0	positive control failed: inconclusive
P-521	jac_b	1.767767	0.707107 [0.0, 2.12132]	4.242641	0.082	100/100	34	within the null distribution
P-521	jac_c	0.883883	0.707107 [0.0, 2.12132]	4.772971	0.4222	100/100	38	within the null distribution
P-521	jac_j	0.707107	0.707107 [0.0, 2.12132]	3.889087	0.5361	100/100	39	within the null distribution
P-521	jac_disc	0.883883	0.707107 [0.0, 2.12132]	4.065864	0.4117	100/100	34	within the null distribution
P-521	depth_b	1.0	1.584963 [0.0, 7.228819]	14.775713	0.7702	100/100	49	within the null distribution
P-521	depth_disc	2.584963	2.584963 [1.0, 7.228819]	13.190751	0.6454	100/100	55	within the null distribution
P-521	hw_b	0.043811	0.657162 [0.043811, 2.234351]	4.074404	1	100/100	45	within the null distribution
P-521	run_b	8.0	9.0 [7.0, 14.0]	20.0	0.8749	100/100	27	within the null distribution
P-521	naf_b	0.375599	0.697542 [0.053657, 2.199939]	3.970621	0.752	100/100	45	within the null distribution
P-521	hw_c	0.964764	0.701646 [0.0, 2.192645]	3.946761	0.348	100/100	52	within the null distribution
P-521	run_c	13.0	9.0 [7.0, 14.0]	23.0	0.0627	100/100	38	within the null distribution
P-521	smooth_b	29.175587	13.549303 [0.0, 42.646906]	83.447502	0.1174	100/100	62	within the null distribution
P-521	smooth_c	1.584963	13.332037 [0.0, 42.365653]	74.937226	0.9269	100/100	48	within the null distribution
P-521	smooth_j	6.266787	13.381678 [0.0, 40.860927]	83.821369	0.7468	100/100	59	within the null distribution
P-521	hw_j	0.131432	0.657162 [0.043811, 2.321972]	4.42489	0.9308	100/100	36	within the null distribution
P-521	run_j	7.0	9.0 [7.0, 14.0]	22.0	0.9858	100/100	35	within the null distribution
P-521	d3	0.0	1.0 [0.0, 4.0]	4.0	1	0/100	0	positive control failed: inconclusive

Sources

- `research/p224_hypothesis_ranking_20261008/README.md` and `PROTOCOL.md`: hypothesis 1, the KS test of `b mod q`, the seed check, the P-224 conductor.
- NIST FIPS 186-4, Appendix D: seeds, `c` and the curve constants; ANSI X9.62 A.3.3.1 (`x962_c_from_seed`).
- Koblitz and Menezes, A Riddle Wrapped in an Enigma; Bernstein et al., How to manipulate curve standards.
- Code: `src/cryptanalysis/b_weak_class_battery.rs`, `src/bin/b_weak_class_battery.rs`.